STRUCTURE OF AN INDECOMPOSABLE INJECTIVE MODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The uniqueness statement for injective hulls. Invoked at the start of the counted proof (source line 788). Gives the first assertion.
- The statement that an injective submodule which is isomorphic to the whole module splits off. Invoked inside the counted proof (source lines 804-806). Makes the image of an injective endomorphism a direct summand.

Source reference. Original lemma lemma-injective-epimorphism: sources/stacks-dualizing/pinned-original-source.tex, source lines 422–422. Navigation pointer only: the counted text refers to the Stacks item lemma-injective-epimorphism, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-injective-hull-unique: sources/stacks-dualizing/pinned-original-source.tex, source lines 719–719. Navigation pointer only: the counted text refers to the Stacks item lemma-injective-hull-unique, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let R be a ring. Let E be an indecomposable injective R-module. Then*

(1) *E is the injective hull of any nonzero submodule of E,*
(2) *the intersection of any two nonzero submodules of E is nonzero,*
(3) *$End_R(E)$ is a noncommutative local ring with maximal ideal those $\varphi : E \to E$ whose kernel is nonzero, and*

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(4) *the set of zerodivisors on E is a prime ideal $\mathfrak{p}$ of R and E is an injective $R_\mathfrak{p}$-module.*

Proof. Part (1) follows from Lemma lemma-injective-hull-unique. Part (2) follows from part (1) and the definition of injective hulls.

Proof of (3). Set $A = \mathrm{End}_R(E)$ and $I = \{\varphi \in A \mid \mathrm{Ker}(\varphi) \neq 0\}$. The statement means that I is a two sided ideal and that any $\varphi \in A$, $\varphi \notin I$ is invertible. Suppose φ and ψ are not injective. Then $\mathrm{Ker}(\varphi) \cap \mathrm{Ker}(\psi)$ is nonzero by (2). Hence $\varphi + \psi \in I$. It follows that I is a two sided ideal. If $\varphi \in A$, $\varphi \notin I$, then $E \cong \varphi(E) \subset E$ is an injective submodule, hence $E = \varphi(E)$ because E is indecomposable.

Proof of (4). Consider the ring map $R \to A$ and let $\mathfrak{p} \subset R$ be the inverse image of the maximal ideal I. Then it is clear that $\mathfrak{p}$ is a prime ideal and that $R \to A$ extends to $R_\mathfrak{p} \to A$. Thus E is an $R_\mathfrak{p}$-module. It follows from Lemma lemma-injective-epimorphism that E is injective as an $R_\mathfrak{p}$-module. $\square$